

Backend Framework Research

1. Node.js (Express / NestJS)

Popular as a server-side JavaScript environment, Node.js is unique in that it is event-driven, non-blocking. Scaling web apps often employ Express.js or NestJS, both of which are quick APIs.

Benefits:

- Handling many requests simultaneously
- Excellent performance on many requests at once
- extensive library ecosystem (npm)
- High compatibility with real-time features

Restrictions:

- Complexity of callbacks when they are disorganized.
- It has to handle CPU-intensive operations very carefully.

The reason to choose Node.js was its ability to handle multiple users concurrently. Handling multiple users is very important when dealing with a thousand students all at once. (Tilkov and Vinoski, 2010).

2. Python FastAPI

FastAPI is the latest version of the Python programming library used for developing APIs with the power of the async

Benefits:

- Fast speed comparable to Node.js
- Automated docs from APIs (OpenAPI/Swagger)
- Great fit if you want integration with AI & ML

Restrictions:

- less extensive ecosystem than Node.js
- require knowledge of asynchronous programming

Given that the proposed system utilizes AI-powered language assistance and adaptive learning intelligence, often developed through Python machine learning

libraries, a detailed study of FastAPI was performed due to its potent potential. (Ramírez, 2020).

3. Justification for Framework Selection

Both FastAPI and Node.js were recognized as good options. FastAPI was identified to be particularly fitting because:

- Ease of integration of AI services
- Pydantic based Clear Request/Response
- Faster creation of well-documented APIs